Step-by-Step Guide to 1NF, 2NF, 3NF, and BCNF with Examples

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1 Introduction

Normalization is the process of organizing a relational database to reduce redundancy and improve data integrity. The most common normal forms are:

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce–Codd Normal Form (BCNF)

We will use the relation:

$$R(A, B, C, D, E, F, G, H)$$

with the functional dependencies:

$$F = \{A \to BC,\ CD \to E,\ B \to D,\ E \to A,\ F \to GH\}$$

2 First Normal Form (1NF)

2.1 Definition

A relation is in 1NF if:

1. All attribute domains contain only atomic (indivisible) values.
2. There are no repeating groups or arrays.

2.2 Checking for 1NF

We check each attribute:

- If any attribute has a set or list of values, it violates 1NF.
- Example: If B contains values like {Math, Physics}, it is not atomic.

2.3 Example

Not in 1NF:

A	B	C	D
1	{x, y}	c1	d1
2	x	c2	d2

In 1NF:

A	B	C	D
1	x	c1	d1
1	y	c1	d1
2	x	c2	d2

3 Second Normal Form (2NF)

3.1 Definition

A relation is in 2NF if:

1. It is already in 1NF.
2. Every non-prime attribute is fully functionally dependent on the **whole** candidate key (no partial dependency).

3.2 Finding Candidate Keys

Using F:

$$A \to BC,\ B \to D,\ CD \to E,\ E \to A,\ F \to GH$$

Closure of AF:

$$AF^{+} = \{A, B, C, D, E, F, G, H\}$$

So (A, F) is a candidate key.

3.3 Partial Dependency Example

If $A \to C$ and (A, F) is a candidate key, C depends only on A (part of key) $\Rightarrow$ violates 2NF.

3.4 Valid in 2NF

Remove partial dependency by decomposing.

4 Third Normal Form (3NF)

4.1 Definition

A relation is in 3NF if:

1. It is already in 2NF.
2. For every functional dependency $\alpha \to \beta$, at least one holds:
   - α is a superkey, or
   - Each attribute in β is prime (part of some candidate key).

4.2 3NF Violation Example

From $B \to D$:

- B is not a superkey.
- D is not prime.
- $\Rightarrow$ violates 3NF.

4.3 Fix

Decompose into (B, D) and (A, B, C, E, F, G, H) while preserving dependencies.

5 Boyce–Codd Normal Form (BCNF)

5.1 Definition

A relation is in BCNF if:

1. For every non-trivial functional dependency $\alpha \to \beta$, α is a superkey.

5.2 BCNF Violation Example

From $E \to A$:

- E is not a superkey.
- $\Rightarrow$ violates BCNF.

5.3 Fix

Decompose into (E, A, B, C, D) and (E, F, G, H) to ensure each left side is a superkey.

6 Summary Table

Normal Form	Violation Type	Fix
1NF	Non-atomic values	Split into multiple rows
2NF	Partial dependency	Remove partial dependencies
3NF	Transitive dependency	Remove non-key determinants
BCNF	Determinant not a superkey	Decompose relation